

Heatwave report card — May 2024 heatwave

REPLAY

SYNTHETIC REPORTS

SIMULATED DELIVERY

Ahmedabad, 48 wards · event 16 May – 30 May 2024 (15 days, peak 26 May) · model estimates, not clinical predictions · event 1 of 2

48 / 48

wards reached Red (mortality risk)

350

Red ward-days · 681 Orange-or-worse

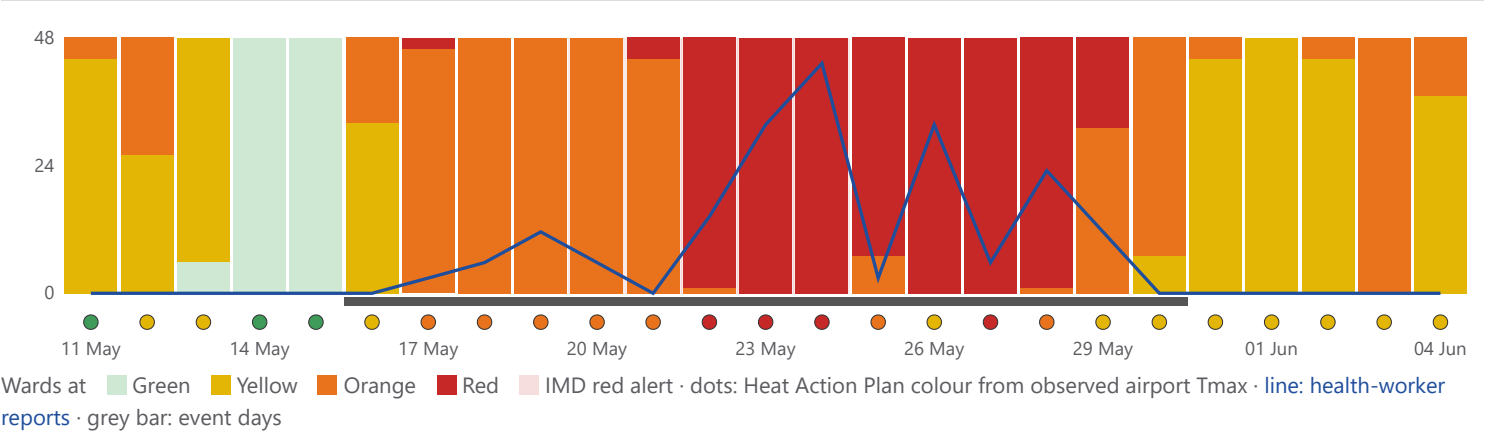
232 of 350

Red ward-days with P(Red) ≥ 40% in forecasts 3–5 days ahead

348 / 350

Red ward-days covered by an approved Red alert

What happened, day by day



Checks against observations

|                                                                                 |               |
|---------------------------------------------------------------------------------|---------------|
| IMD red-alert days (20 May–24 May) with a ward at Red                           | 4 of 5        |
| Days of Orange-or-worse warning before IMD's red alert                          | 4             |
| Plan Orange/Red days (observed Tmax) the model also rated Orange+ (median ward) | 11 of 11      |
| Missed: plan Orange/Red, model below                                            | 0             |
| Over-warning: model Orange+, plan Yellow or none                                | 4 of 14       |
| Same level as the plan                                                          | 18 of 25 days |

References: IMD red alert for Ahmedabad 20–24 May 2024; Heat Action Plan colour from observed Tmax at the airport (IMD 42647). Health outcomes are city-level only (69 heatstroke cases in May 2024).

Health-worker reports

66 reports in the window (synthetic demo data). 3 ward-days flagged for more cases than the model's share.

| Ward       | Day    | Reports | Expected | MRI  |
|------------|--------|---------|----------|------|
| Vatva      | 23 May | 6       | 0.19     | 89.7 |
| Maktampura | 24 May | 3       | 0.25     | 85.3 |
| Vatva      | 24 May | 7       | 0.26     | 90.9 |

Alerts and actions

27 alert drafts: 20 dispatched, 7 draft.  
Messages: sms 54 simulated; voice 20 simulated; whatsapp 20 simulated.

| Level  | Action     | Alerts | Days          | Officer                  |
|--------|------------|--------|---------------|--------------------------|
| Red    | approved   | 9      | 21 May–29 May | Demo officer (simulated) |
| Red    | dispatched | 9      | 21 May–29 May | Demo officer (simulated) |
| Orange | approved   | 11     | 16 May–30 May | Demo officer (simulated) |
| Orange | dispatched | 11     | 16 May–30 May | Demo officer (simulated) |

Every decision is in the audit log with officer name and time.

Most affected wards

| Ward                | Red days | Peak MRI | Reports / exp. |
|---------------------|----------|----------|----------------|
| Indrapuri           | 10       | 100      | 0 / 1.3        |
| Baherampura         | 10       | 100      | 0 / 2.0        |
| Odhav               | 9        | 99       | 4 / 1.9        |
| Asarwa              | 9        | 99       | 0 / 0.9        |
| Bhaipura Hatkeshwar | 8        | 96       | 0 / 0.8        |
| Amraiwadi           | 8        | 96       | 2 / 1.3        |
| Vasna               | 8        | 96       | 0 / 1.3        |

|              |   |    |         |
|--------------|---|----|---------|
| India Colony | 8 | 96 | 0 / 1.6 |
| Shahpur      | 8 | 96 | 2 / 1.1 |
| Lambha       | 8 | 95 | 2 / 2.9 |

## Limitations

- Model estimates, not clinical predictions. Health outcomes for this event are known only city-wide.
- Early warning in a replay uses archived deterministic forecasts from three models (9 members), not the live 122-member ensemble.
- Sheet-roof shares are census-based estimates; elderly, children and outdoor-worker indicators are held at the city midpoint.
- Report flags are prompts to verify, not confirmed clusters: at  $p < 0.01$  over hundreds of ward-days, about one flag can occur by chance.
- Health-worker reports in this card are SYNTHETIC demo data.
- Alert deliveries were SIMULATED: nothing was sent to real phones.

## Appendix — all wards

| Ward                | Red days | Orange+ days | First Red | Red days alerted | Peak MRI | PVI | Reports | Expected |
|---------------------|----------|--------------|-----------|------------------|----------|-----|---------|----------|
| Indrapuri           | 10       | 15           | 17 May    | 9                | 100      | 58  | 0       | 1.3      |
| Baherampura         | 10       | 15           | 17 May    | 9                | 100      | 61  | 0       | 2.0      |
| Odhav               | 9        | 15           | 21 May    | 9                | 99       | 59  | 4       | 1.9      |
| Asarwa              | 9        | 15           | 21 May    | 9                | 99       | 57  | 0       | 0.9      |
| Bhaipura Hatkeshwar | 8        | 15           | 22 May    | 8                | 96       | 55  | 0       | 0.8      |
| Amraiwadi           | 8        | 15           | 22 May    | 8                | 96       | 55  | 2       | 1.3      |
| Vasna               | 8        | 15           | 22 May    | 8                | 96       | 55  | 0       | 1.3      |
| India Colony        | 8        | 15           | 22 May    | 8                | 96       | 55  | 0       | 1.6      |
| Shahpur             | 8        | 15           | 22 May    | 8                | 96       | 52  | 2       | 1.1      |
| Lambha              | 8        | 14           | 22 May    | 8                | 95       | 58  | 2       | 2.9      |
| Thakkarbapanagar    | 8        | 15           | 22 May    | 8                | 95       | 54  | 1       | 1.0      |
| Ranip               | 8        | 15           | 22 May    | 8                | 95       | 54  | 0       | 1.2      |
| Sabarmati           | 8        | 15           | 22 May    | 8                | 95       | 54  | 1       | 1.4      |
| Danilimda           | 8        | 15           | 22 May    | 8                | 94       | 52  | 2       | 1.1      |
| New Wadaj           | 8        | 15           | 22 May    | 8                | 94       | 49  | 0       | 1.0      |
| Saraspur-Rakhiyal   | 8        | 15           | 22 May    | 8                | 94       | 52  | 0       | 1.1      |
| Ghatlodia           | 8        | 15           | 22 May    | 8                | 93       | 52  | 0       | 1.0      |
| Naroda              | 7        | 14           | 22 May    | 7                | 93       | 53  | 2       | 2.6      |
| <b>Vatva</b>        | 7        | 14           | 22 May    | 7                | 92       | 52  | 13      | 1.2      |
| S.P.Stadium         | 7        | 14           | 22 May    | 7                | 92       | 50  | 0       | 1.4      |
| Gomtipur            | 7        | 14           | 22 May    | 7                | 92       | 51  | 2       | 1.2      |
| Chandlodia          | 7        | 14           | 22 May    | 7                | 92       | 52  | 1       | 0.6      |
| Jamalpur            | 7        | 14           | 22 May    | 7                | 92       | 49  | 0       | 0.9      |
| Maninagar           | 7        | 14           | 22 May    | 7                | 92       | 49  | 0       | 1.7      |
| Vejalpur            | 7        | 14           | 22 May    | 7                | 92       | 50  | 2       | 1.0      |
| Sardarnagar         | 7        | 14           | 22 May    | 7                | 91       | 51  | 2       | 1.4      |
| Nikol               | 7        | 14           | 22 May    | 7                | 91       | 49  | 0       | 2.3      |
| Khadia              | 7        | 14           | 22 May    | 7                | 91       | 48  | 3       | 1.2      |
| Viratnagar          | 7        | 14           | 22 May    | 7                | 91       | 49  | 1       | 1.4      |
| Isanpur             | 7        | 14           | 22 May    | 7                | 91       | 48  | 0       | 1.4      |
| Shahibag            | 7        | 14           | 22 May    | 7                | 90       | 48  | 2       | 1.4      |
| Gota                | 7        | 14           | 22 May    | 7                | 90       | 50  | 3       | 1.1      |
| Sarkhej             | 7        | 14           | 22 May    | 7                | 90       | 51  | 2       | 2.3      |
| Ramol Hathijan      | 7        | 14           | 22 May    | 7                | 90       | 51  | 2       | 2.3      |
| Bodakdev            | 7        | 14           | 22 May    | 7                | 90       | 48  | 2       | 1.1      |
| Vastral             | 7        | 14           | 22 May    | 7                | 90       | 49  | 1       | 0.9      |
| Thaltej             | 7        | 14           | 22 May    | 7                | 90       | 50  | 1       | 1.0      |
| Khokhra             | 7        | 14           | 22 May    | 7                | 90       | 47  | 0       | 1.4      |

|               |   |    |        |   |    |    |   |     |
|---------------|---|----|--------|---|----|----|---|-----|
| Saijpur Bogha | 7 | 14 | 22 May | 7 | 88 | 45 | 2 | 1.0 |
| Navrangpura   | 7 | 14 | 22 May | 7 | 88 | 45 | 2 | 3.7 |
| Bapunagar     | 7 | 14 | 22 May | 7 | 87 | 44 | 1 | 0.9 |
| Maktampura    | 6 | 13 | 22 May | 6 | 87 | 48 | 3 | 1.1 |
| Kubernagr     | 6 | 13 | 22 May | 6 | 86 | 42 | 0 | 0.5 |
| Chandkheda    | 6 | 13 | 22 May | 6 | 85 | 42 | 0 | 0.4 |
| Paldi         | 6 | 13 | 22 May | 6 | 84 | 40 | 0 | 1.7 |
| Dariyapur     | 6 | 13 | 22 May | 6 | 84 | 39 | 1 | 0.6 |
| Naranpura     | 6 | 13 | 22 May | 6 | 84 | 39 | 3 | 1.5 |
| Jodhpur       | 4 | 13 | 23 May | 4 | 83 | 40 | 1 | 2.1 |

Early-warning source: Weather: ERA5 reanalysis. Probabilities: forecasts issued 3-5 days ahead by ECMWF, GFS and ICON (9 members), i.e. what the system would have shown 3 days before each day.